

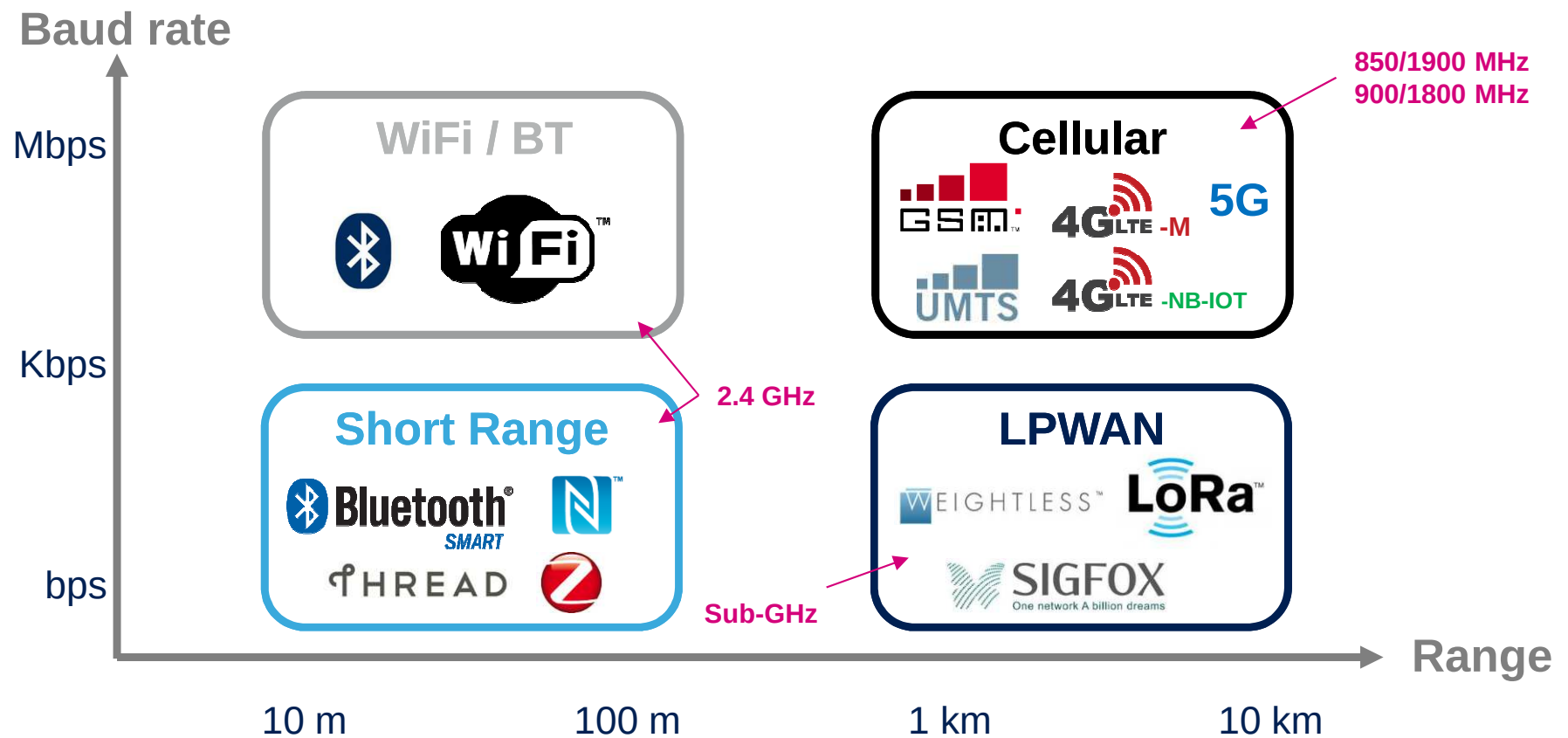
LoRa & LoRaWAN



Hands-on

- Key learning
 - LoRa technology modulation
 - LoRa PHY layer
 - End node classes
 - Security
 - End node network activation

Communication Technologies - Overview



ISM World wide regulation

Output power vs Duty Cycle

- Europe:

- In the EU the ISM regulation is based 868-869.2MHz Frequency Bands

Frequency Bands	ERP	Duty Cycle	Channel Bandwidth
868 – 868.6MHz	+14 dBm	<1%	No limits
868.7 – 869.2 MHz	+14 dBm	< 0.1%	No limits

- Duty cycle: If a given device talk about 1s then this same device have to keep silent for the next 99 second.

- Americas:

- In the USA the ISM regulation is based 902-928MHz Frequency Bands
 - if the 20 dB bandwidth of the hopping channel is less than 250 kHz, the system shall use at least 50 hopping frequencies and the average time of occupancy on any frequency shall not be greater than 0.4 seconds within a 20 second period with 1 Watt output power permitted
 - if the 20 dB bandwidth of the hopping channel is 250 kHz or greater, the system shall use at least 25 hopping frequencies and the average time of occupancy on any frequency shall not be greater than 0.4 seconds within a 10 second period with 0.25 Watts output power permitted
 - The maximum allowed 20 dB bandwidth of the hopping channel is 500 kHz.

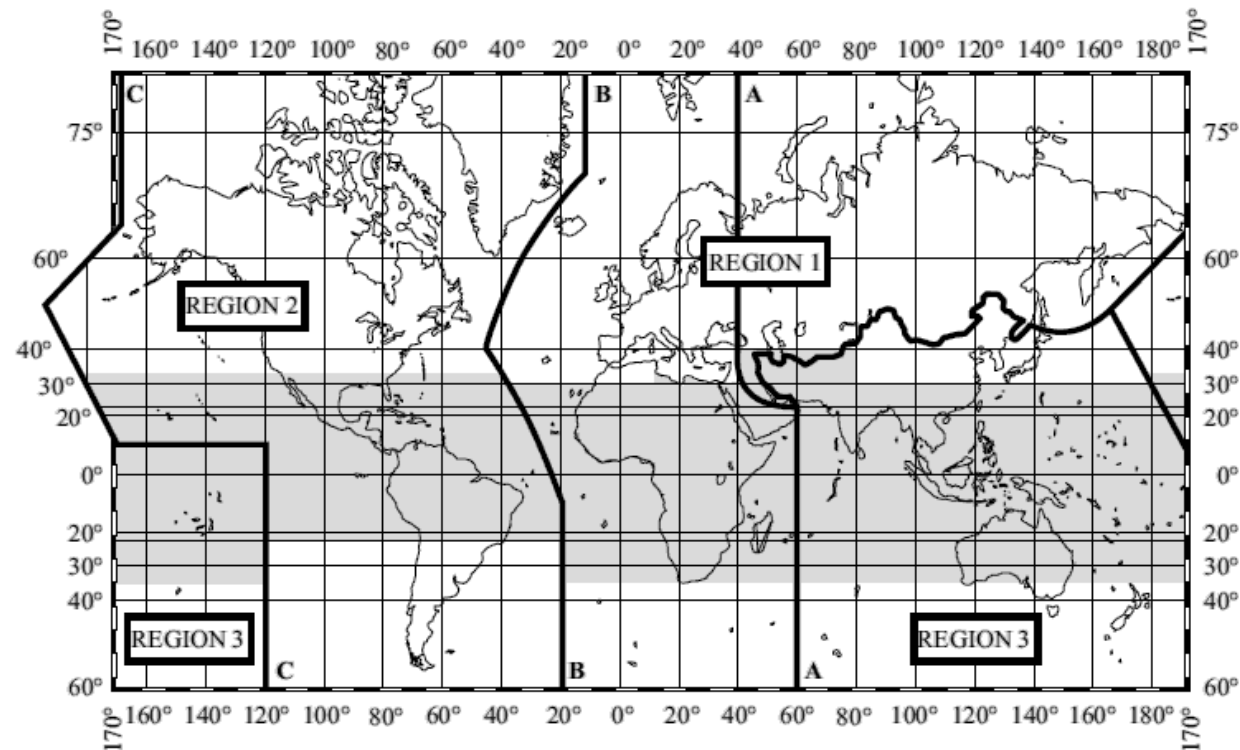
- Asia:

- Mix between, USA, EU or own Countries ISM bands regulation

ISM World wide regulation

Output power vs Duty Cycle

Countries	Frequency Band Review	Max Output Power
EU	868MHz	14dBm
USA	915MHz	20dBm
Korea		14dBm
Japan		
Malaysia	862MHz to 875MHz	20dBm
Philippines	868MHz	
Vietnam	862MHz to 875MHz 315/433/.. TBC	
India	865MHz - 867MHz	
Singapore	922MHz	
Thailand	922MHz/868MHz TBC	
Indonesia	922MHz	
ANZ	915MHz to 928MHz	
Taiwan	920MHz to 925MHz	
China	470MHz to 510MHz	17dbm



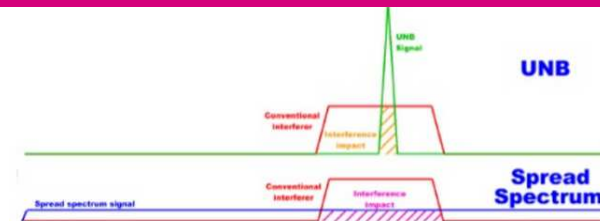
Sigfox and LoRa® - Overview

2 different technology to support IoT

1. Ultra Narrow Band (UNB) →



2. Spread Spectrum (SS) →

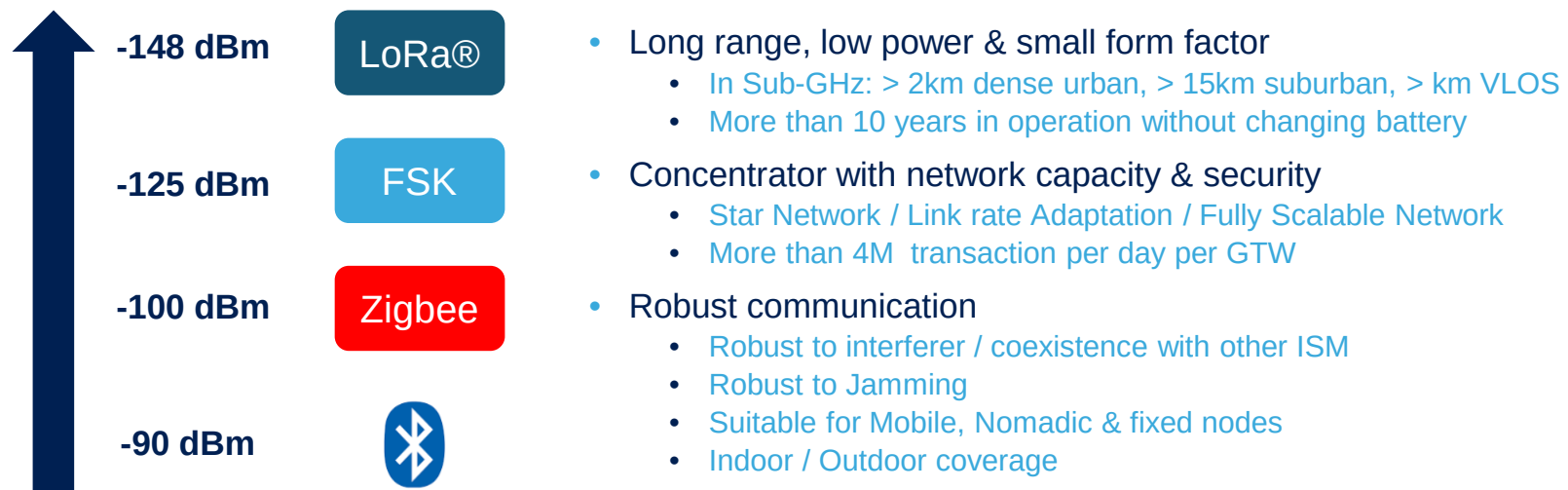


	Sigfox	LoRa®
Modulation	UNB	DSS-like
Throughput	100bps	300bps to 50Kbps
Payload	12bytes	64bytes
Link Adaptation	No (BPSK)	VSF (SF7-SF12)
BW	100Hz	125KHz
DutyCycle Limited	Yes	Yes
Channel Hopping	Yes	Yes
Best Sensitivity (dBm)	-142	-142
Bi-Directional	4 message maximum /day	Yes
Battery Life	10years	10Years
Localization	No	Yes (30m)
Encryption	AES-128	AES-128
SDR	Yes	N/A
Benefits	Robustness to RF coexistence / Multi Radio vendor Low cost module / Network deployed in EU	2 to 3x longer Range / Less sensitive to noise and environment True Bi-directional / Private & Public network possible WW massive network deployment
Drawbacks	Limited baud rate (->limited application) /Limitation in the USA Not true Bi-dir / Only 1 network possible	Single radio provider

What is LoRaTM ?

What is LoRa™ ?

1. A Sub-GHz wireless technology enabling low data rate communications over long distances
2. Is targeting M2M and Internet of Things, IoT applications
3. LoRa® technology is a solution providing a WAN capability, using a MAC protocol named LoRaWAN



LoRaWAN™ Differentiation and Benefits



True location

- Indoor and outdoor
- Accurate



Bidirectional

- Bidirectional
- Scalable capacity
- Broadcast



Global mobility

- True mobility
- Seamless
- Roaming



Security

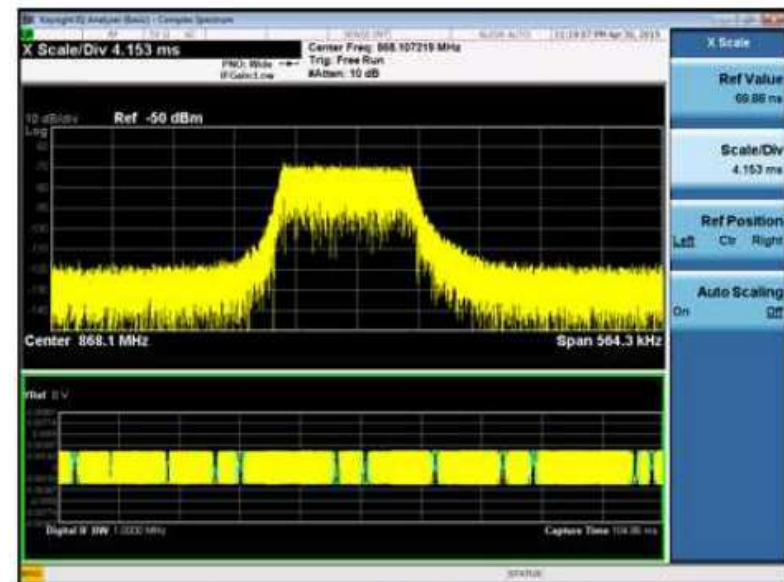
- Unique ID
- Application
- Network



LoRa™ Technology Modulation and LoRaWAN™ Network Protocol

LoRa™ Technology Modulation

- LoRa™ technology is based on the Spread Spectrum Technology
- LoRa™ have been developed by Cycleo then acquired by Semtech
- It is a Chirped Frequency Modulation
- LoRa™ Spread Spectrum technology means:



LoRa™ Technology Modulation

Let's start from the ground (1/2)

- Spread Factor (SF): each bit of information is encoded as multiple chips, chip rate R_c and bit rate R_b is given by: $R_c = 2^{SF} * R_b$
- Bandwidth (BW): frequency of the spread signal in kHz. LoRaWAN supports 125, 250 and 500kHz. For a given SF, a narrower BW (ex: 125kHz) will increased receive sensitivity and so the TOA.
- Time On Air (TOA): time requested to transmit data over the air for a given SF, $T_{sym} = 2^{SF} / BW$.
The higher is SF, the longer is TOA; the wider is BW, the shorter is TOA
- Bit Rate (BR): number of bits per second on the TX phase
- Forward Error Correction (FEC) and Coding Rate (CR): ratio of byte used to perform the correction. Ex: a CR 4/5 means 20% of the payload is used as error correction code (ECC like) named FEC.
The higher percentage of FEC in payload, the longer is TOA
- Range: distance for given setting. The value is dBm rather than km or meters
- Receiver Sensitivity (dBm): the lower the more sensitive is radio receiver.
The wider is BW, the lower receiver sensitivity due to integration of additional noise power in the channel.

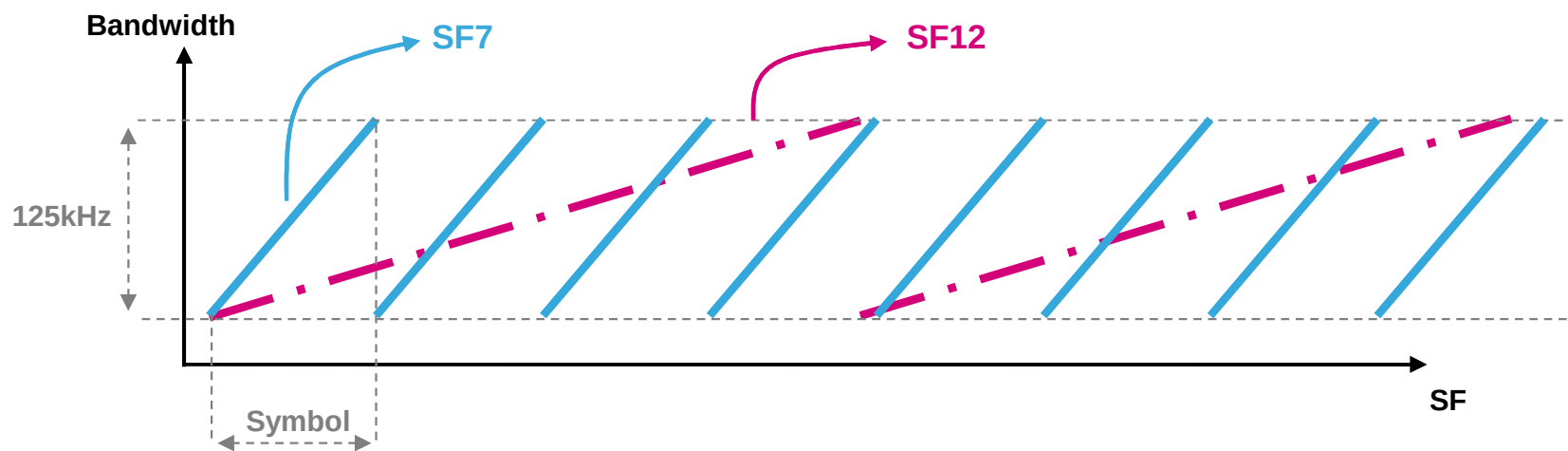
LoRa™ Technology Modulation

Let's start from the ground (2/2)

- Data Rate (DR): LoRa™ modulation (could be FSK with an impact on the range)
- Adaptive Data Rate (ADR): LoRaWAN™ can adapt DR and RF output power of each device to
 - Maximize battery life
 - Maximize network capability
 - Set the best setting for fastest DR

LoRa™ Technology Modulation

Understanding the impact of different parameters

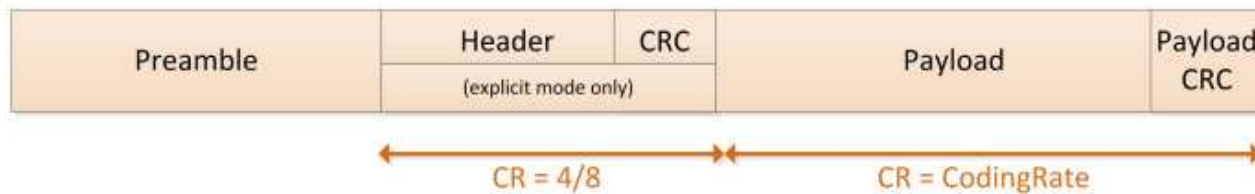


Bandwidth	SF	Symbol	Nb bit per symbol	SNR	Sensitivity	BR	TOA
125kHz	SF7	7ms	1	-7.5dBm	-118	5kbps	10ms
	SF12	32ms	12	-20dBm	-136	300bps	1.5s

As a conclusion: the higher is the SF, the longer is the TOA, the less bytes are sent
BUT the longer is the range(distance) achieved.

LORA PHY (2)

- PHY PACKET STRUCTURE



- preamble is used to synchronize receiver with the incoming signal.
 - 0x34 so 8 preamble symbols
- header provides information on the payload with codingRate=4/8:
 - The payload length in bytes.
 - The forward error correction code rate
 - The presence of an optional 16-bits CRC for the payload.
- Payload contains the MacHeader and MacPayload and a MAC integrity code
 - Compatible with 802.15.4 recommendations
- Optional CRC in uplink message

LORA PHY

- Spread spectrum: Chirp cyclic shift modulation, information in shift
- LORA modulation is patented.
 - The modem is delivered as digital IP for integration
- Several data-rate are available 18bps to 37.5kBps
 - By choosing Spreading Factors (chips/bits), RF bandwidth and Coding Rates

- Example of LORA performance

Bandwidth (kHz)	Spreading Factor	Coding rate	Nominal Rb (bps)	Sensitivity indication (dBm)	Frequency Reference
10.4	6	4/5	782	-131	TCXO
	12	4/5	24	-147	
20.8	6	4/5	1562	-128	
	12	4/5	49	-144	
62.5	6	4/5	4688	-121	XTAL
	12	4/5	146	-139	
125	6	4/5	9380	-118	
	12	4/5	293	-136	

Medium Access (2)

• MAC PACKET STRUCTURE

- MHDR: mac message type (Join Request/Accept message OR Data up/down message)
 - Data messages are used to transfer both MAC commands and application data
- FHDR: contains the DevAddr, adaptative rate control, frame counter, Acknowledgement, mac commands if any
- Fport
 - =0: FRMPayload contains MAC commands only, FRMPayload encrypted with NwkSKey
 - otherwise (1..223) are application specific FRMPayload encrypted with AppSKey
- The message integrity code (MIC) is calculated over all the fields in the message (AES).



Figure 5: Radio PHY structure (CRC* is only available on uplink messages)



MACPayload:

Figure 7: MAC payload structure

FHDR:

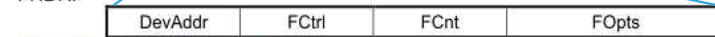


Figure 8: Frame header structure

Bit#	7	6	5	4	[3..0]
FCtrl bits	ADR	ADRACKReq	ACK	FPending	FOptsLen

CID	Command	Transmitted by End-device	Gateway	Short Description
0x02	LinkCheckReq	x		Used by an end-device to validate its connectivity to a network.
0x02	LinkCheckAns		x	Answer to LinkCheckReq command. Contains the received signal power estimation indicating to the end-device the quality of reception (link margin).
0x03	LinkADRRReq		x	Requests the end-device to change data rate, transmit power, repetition rate or channel.
0x03	LinkADRRAns	x		Acknowledges the LinkRateReq.
0x04	DutyCycleReq		x	Sets the maximum aggregated transmit duty-cycle of a device
0x04	DutyCycleAns	x		Acknowledges a DutyCycleReq command
0x05	RXParamSetupReq		x	Sets the reception slots parameters
0x05	RXParamSetupAns	x		Acknowledges a RXSetupReq command
0x06	DevStatusReq		x	Requests the status of the end-device
0x06	DevStatusAns	x		Returns the status of the end-device, namely its battery level and its demodulation margin
0x07	NewChannelReq		x	Creates or modifies the definition of a radio channel
0x07	NewChannelAns	x		Acknowledges a NewChannelReq command
0x08	RXTimingSetupReq		x	Sets the timing of the of the reception slots
0x08	RXTimingSetupAns	x		Acknowledge RXTimingSetupReq command
0x80 to 0xFF	Proprietary	x	x	Reserved for proprietary network command extensions

Table 4: MAC commands

Medium Access

- The current LoRaWAN specification exclusively uses duty-cycled limited transmissions to comply with the ETSI regulations.
 - The LoRaWAN enforces a per sub-band duty-cycle (1%) limitation. Each time a frame is transmitted in a given sub-band, the time of emission and the on-air duration of the frame are recorded for this sub-band. The same sub-band cannot be used again during the next T_{off} seconds. During the unavailable time of a given sub-band, the device may still be able to transmit on another sub-band.
 - Tx during 0.5sec, subband unavailable during 49.5sec
 - Other subbands can be tried. If all subbands are unavailable due to DC limitation, device has to wait...

$$T_{off_subband} = \frac{TimeOnAir}{DutyCycle_{subband}} - TimeOnAir$$

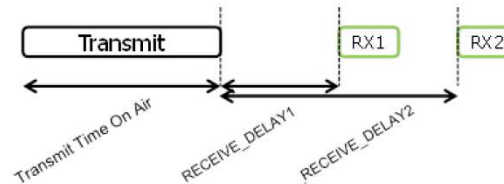
LoRaWAN™ Devices Classes

3 classes to cover all the use cases

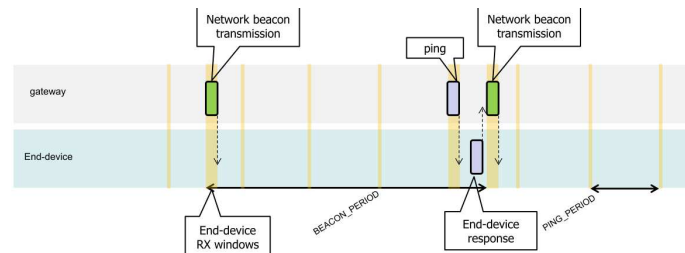
Class name	Intended usage	
A (« all »)	Battery powered sensors (or actuators with no latency constraint) Most energy efficient communication class. Must be supported by all devices	<i>Mainly uplink with two potential downlink slots after each uplink</i>
B (« beacon »)	Battery powered actuators Energy efficient communication class for latency controlled downlink. Based on slotted communication synchronized with a network beacon.	<i>Programmed downlink slots to allow control within certain latency limits</i>
C (« continuous »)	Mains powered actuators Devices which can afford to listen continuously. No latency for downlink communication.	<i>Lowest latency command and control for less power critical devices.</i>

Medium Access

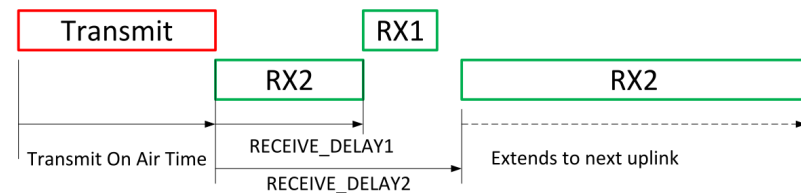
- LORA CLASS A: when sensor has info (random), it reports it to network. Network can respond during Rx1 or Rx2 receive window



- CLASS B: adds a synchronized reception window on the end-device. LORA end-device joins always in class A and might switch to class B upon application layer request.

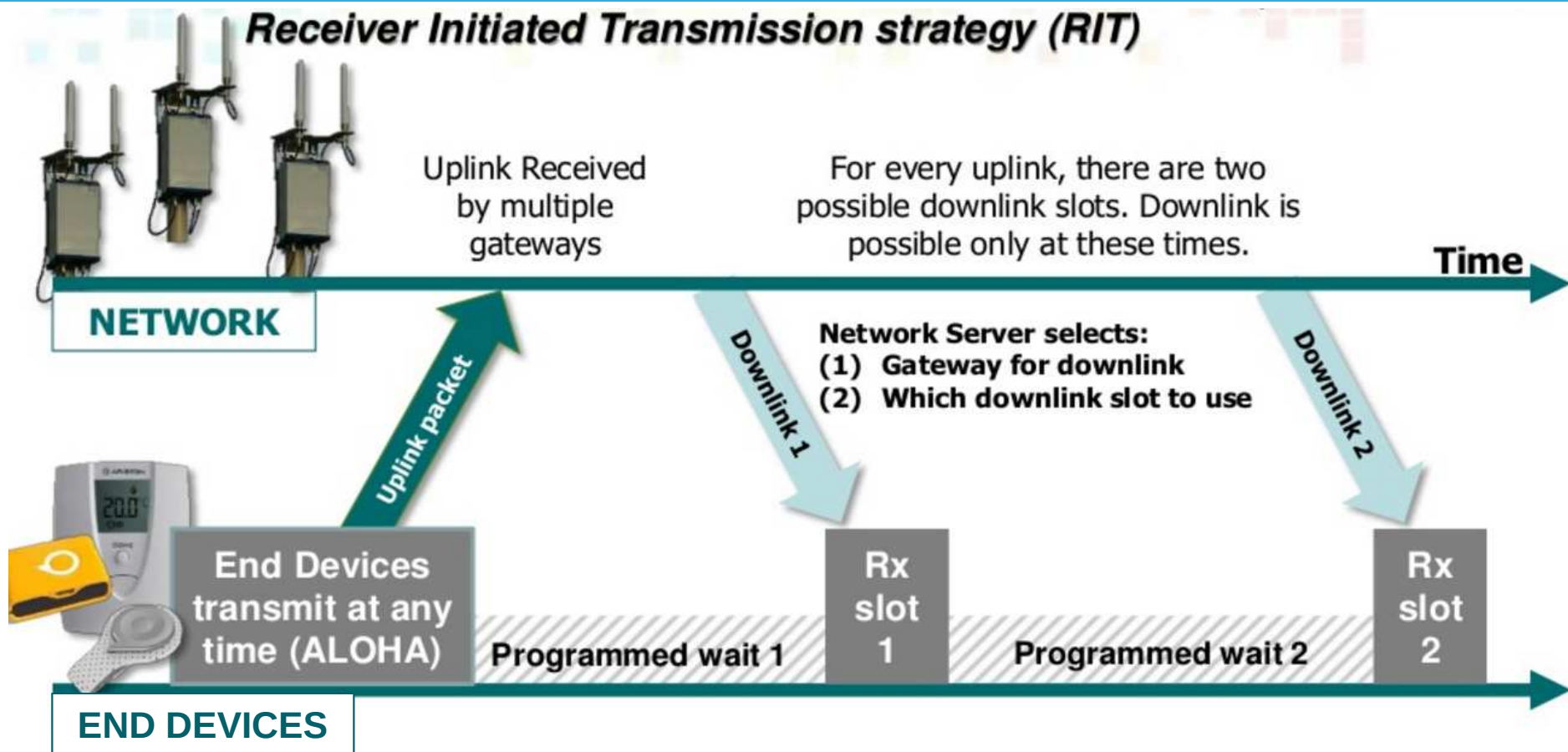


- Class C devices implement the same two receive windows as Class A devices, but they do not close RX2 window until they need to send again. (It is up to the application on server side to know it manages Class C nodes based on the contract passed during the join procedure.)



LoRaWAN™ Devices Classes

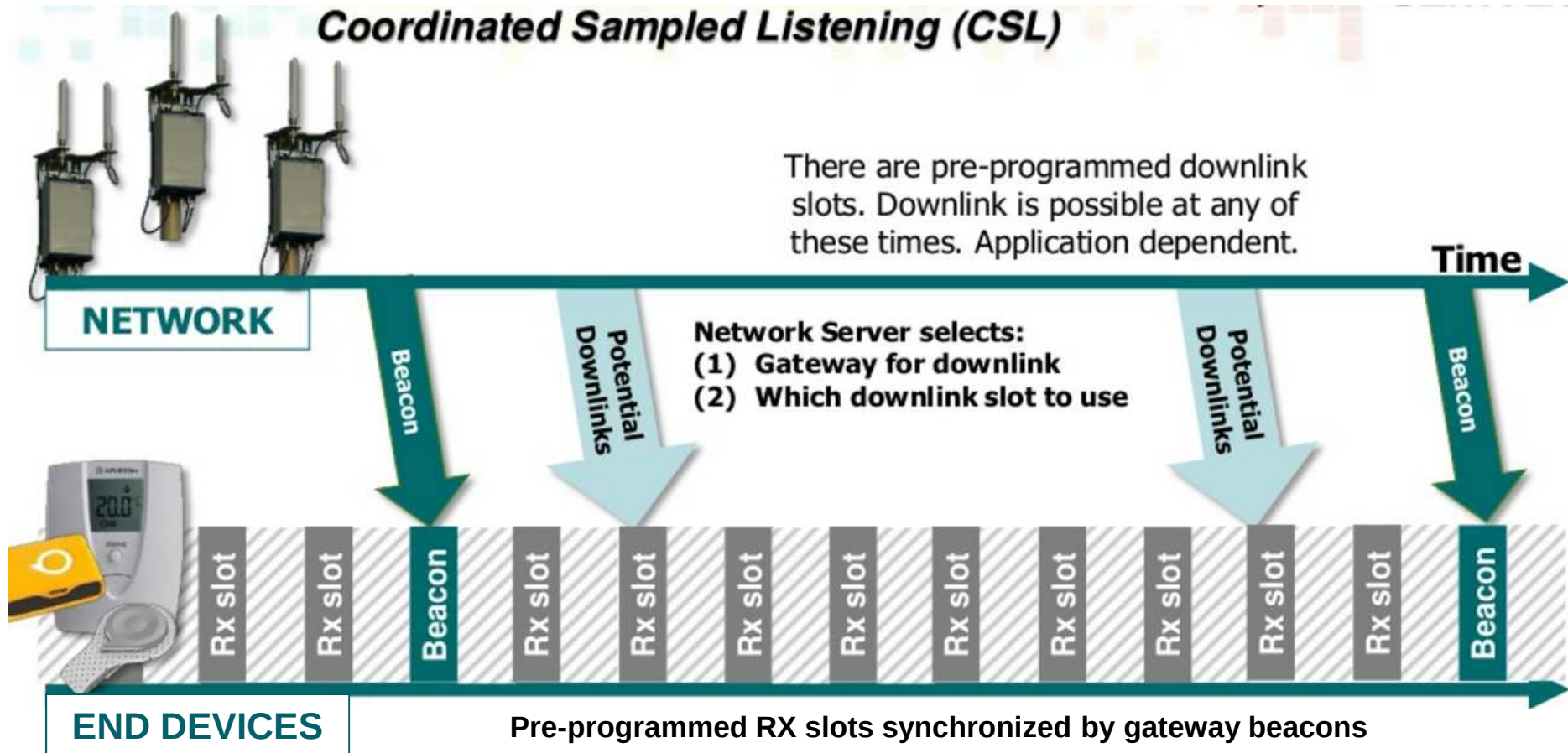
Class A – Bidirectional Communication



LoRaWAN™ Devices Classes

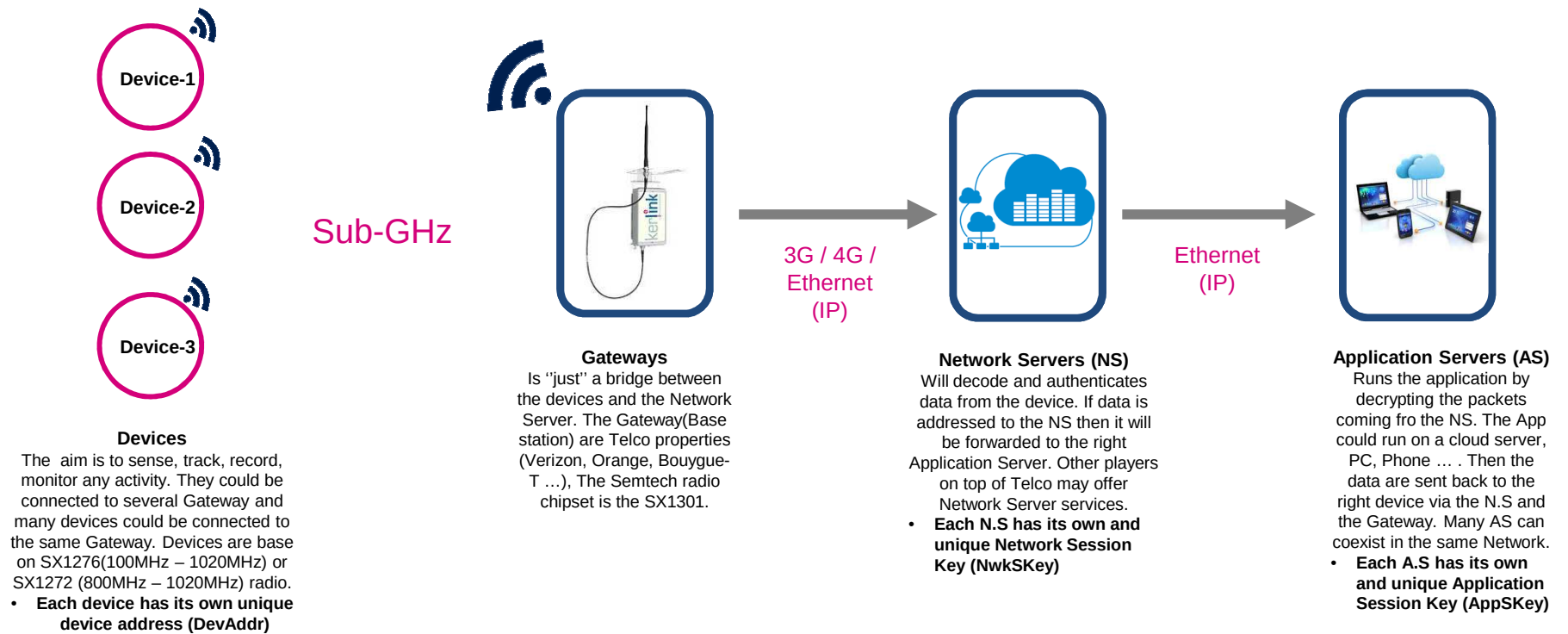
Class B – Bidirectional Communication

Coordinated Sampled Listening (CSL)



LoRaWAN™ Network Protocol

Network Topology Overview

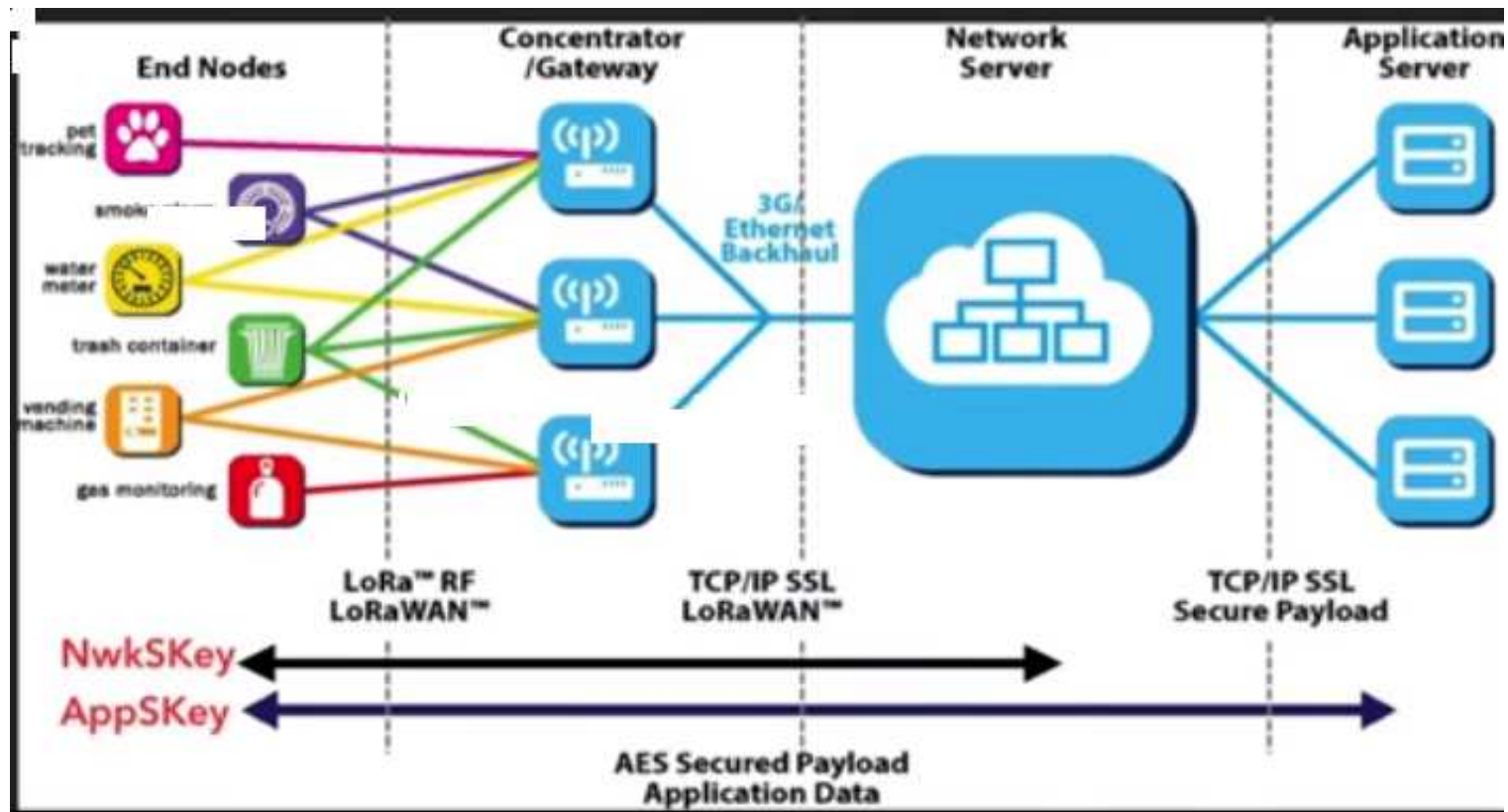


LoRaWAN™ - Security

A native AES 128-bit security network protocol

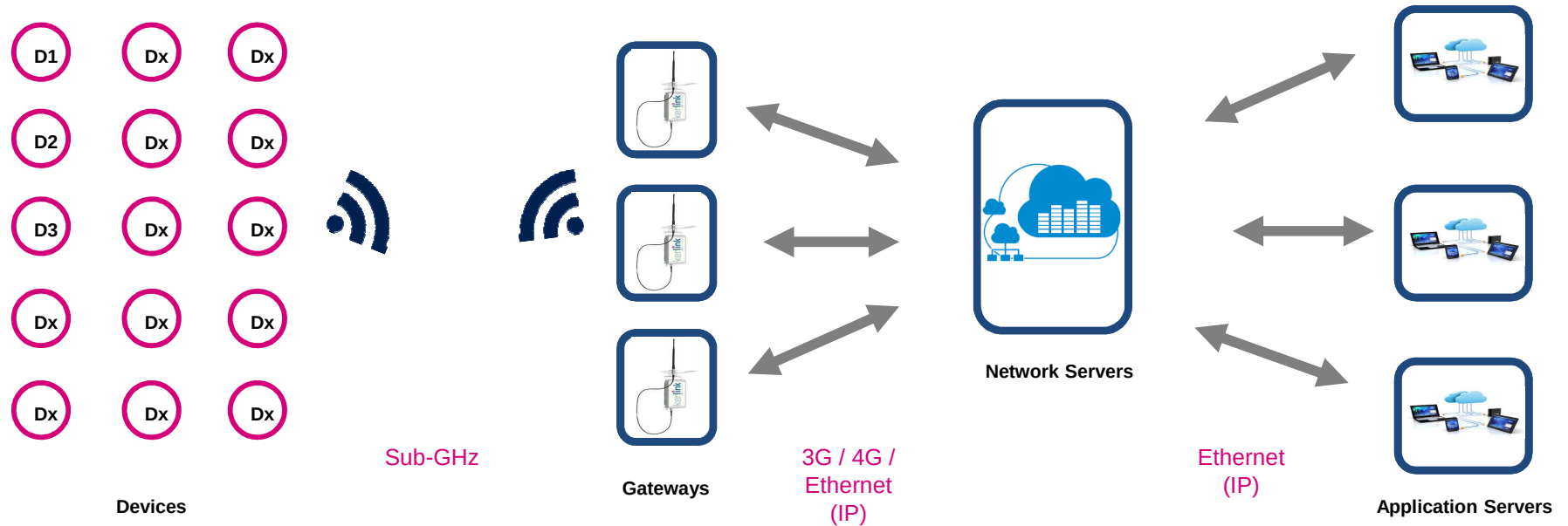
- Device Address (DevAddr) is a 32-bit identifier
 - Unique within the network
 - Available in each data frame and shared between end-device, N.S and A.S
- Network Session Key (NwkSKey) is a 128-bit AES encryption key
 - Unique per end-device and shared between end-device and N.S
 - It allow message integrity communication between end-device and N.S
- Application Session Key (AppSKey) is a 128-bit AES encryption key
 - Unique per end-device and shared between end-device and A.S
 - It is used to encrypt / decrypt A.S server messages to the end-device
- To increase end-device authentication and security a secure can be added in the Device

LoRa network and Security



LoRa™ Network Protocol

Real Network Topology Deployed



LoRaWAN™ – Joining Process

2 activation methods exist to join an existing network

1. OTTA: Over the Air Activation = Over the air handshaking

- End-device send a Join Request (J.S) message to A.S including DevEUI, AppEUI, AppKey
 - DevEUI (Global unique end-device identifier) will allow the network to identify the device
 - AppEUI to identify which A.S the end-device is talking to
 - AppKey to identify which application on the A.S the end-device is referring to
- End device receives the Join Accept from the A.S and proceed with
 - Join Accept (authenticates and decrypt the Join Accept)
 - Extract and store DevAddr (the 32-bit device address)
 - Derives the 2 security: Network session Key (NwkSKey) and Application Session Key (AppSKey)

2. ABP: Activation By Personalization (no Over the air handshaking)

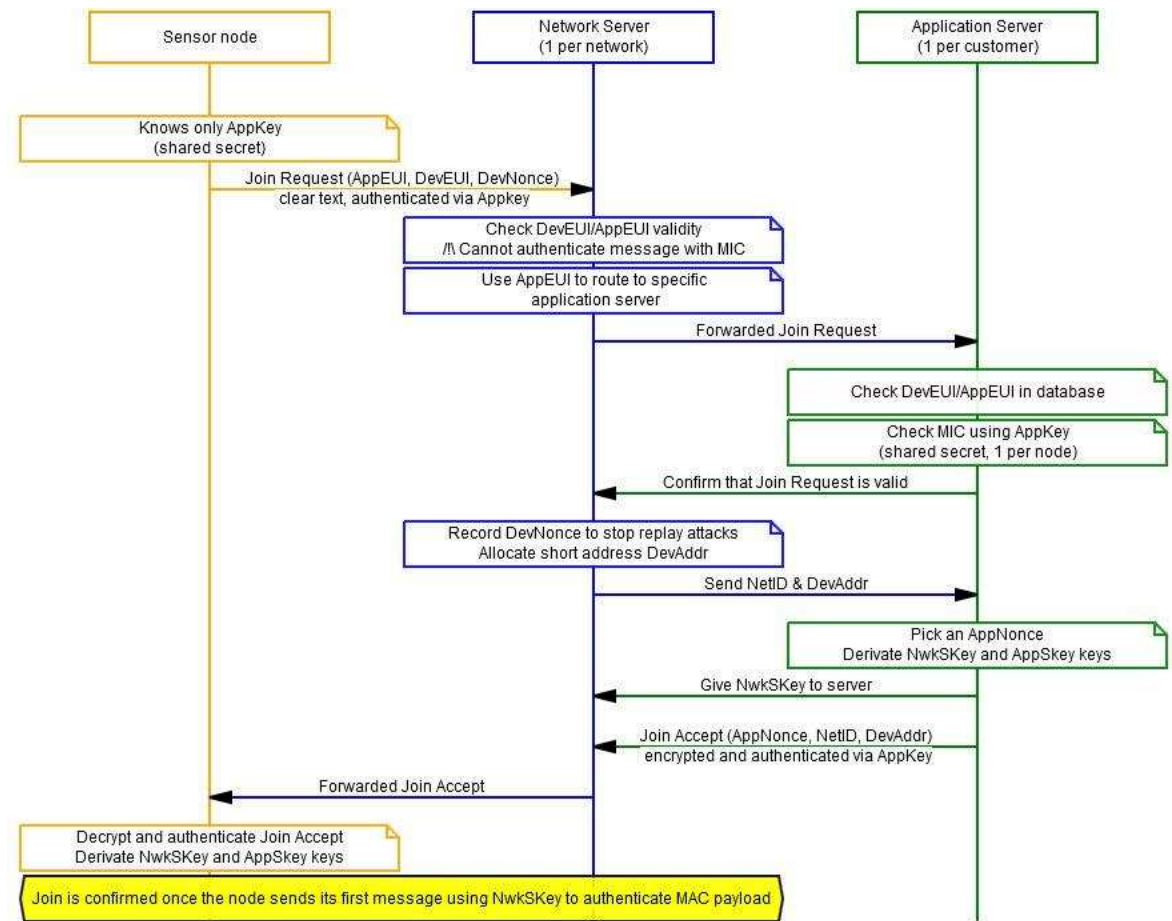
- The Device ready out-of-the-box to talk on the network
- The DevAddr, NwkSKey and AppSKey are programmed at factory level (no Over the air handshaking)

Over The Air Activation

- In OTAA, a join procedure is required before a Lora Object can send any data to the network.
- The device is personalized with an 128 bit AES key (appkey) and two identifiers (appEui and devEui)
 - appkey is known by both the object and the application server running over the LoRa network.

DevNonce is a random value. For each end-device, the network server keeps track of a certain number of **DevNonce** values used by the end-device in the past, and ignores join requests with any of these **DevNonce** values from that end-device.

This mechanism prevents replay attacks by sending previously recorded join-request messages with the intention of disconnecting the respective end-device from the network.



Activation by Personalisation

- In ABP, the device is already personalized with a DevAddr, NwSKey and an AppSKey. No Join procedure is required.
- Since the device is already registered in the LoRa Network, it can directly send data to it.
- However, roaming is not possible.

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